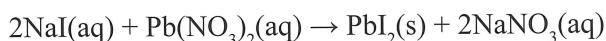


Set 27: Ionic equations

Notes

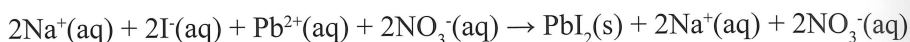
3. Write a balanced ionic equation showing the products formed when a solution of sodium iodide is added to a solution of lead(II) nitrate.

(a) Write a balanced formula equation:

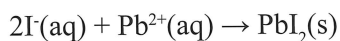


Note: Lead(II) iodide is an insoluble salt whereas sodium nitrate is a soluble salt.

(b) Write relevant species as ions:



(c) Omit spectator ions and write the balanced ionic equation:



Set 27: Exercises

Write balanced ionic equations for the following reactions:

1. The dissolving of potassium chloride.
2. The dissolving of barium nitrate.
3. The dissolving of sodium hydroxide.
4. Dilute nitric acid added to a solution of potassium hydroxide.
5. Zinc oxide powder added to sulfuric acid solution.
6. Dilute nitric acid being added to solid calcium carbonate.
7. A piece of clean lead added to a solution of silver nitrate.
8. The addition of a silver nitrate solution to a solution containing magnesium chloride.
9. Carbon dioxide gas being bubbled through limewater solution.
10. Phosphoric acid added to solid silver carbonate.
11. Hydrogen sulfide gas bubbled through silver nitrate solution.
12. Mixing of nitric acid and sodium carbonate solutions.
13. Sodium iodide solution added to lead(II) acetate solution.
14. The addition of excess sodium hydroxide solution to a suspension of aluminium hydroxide in water to produce a sodium tetrahydroxyaluminate ($\text{Na}[\text{Al}(\text{OH})_4]$) solution.